

An Agent-Based Digital Twin Framework for Predicting and Early Detection of Decreasing Frozen Fruit Quality

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Abstract—Frozen fruit quality degrades silently. Ice crystal formation, freeze-thaw cycles, and freezer burn damage the fruit from the inside out long before any visible signs appear on the surface. By the time cold-chain managers notice something is wrong, the damage is already done and the loss is unavoidable. Current inspection methods rely on periodic lab testing or surface-level visual checks, both of which are too slow, too expensive, and entirely reactive. We propose an agent-based digital twin framework that combines fruit image analysis with real-time sensor data temperature, humidity, and CO₂ to continuously monitor, forecast, and explain frozen fruit quality decline throughout cold-chain storage. When quality crosses a warning threshold, an intelligent agent automatically triggers alerts and recommends corrective actions, turning a reactive process into a proactive one.

The proposed system comprises five modules: (1) a convolutional neural network (CNN) classifier based on EfficientNet-B0 with transfer learning and Gradient-weighted Class Activation Mapping (Grad-CAM) explainability; (2) a Bidirectional Long Short-Term Memory (BiLSTM) network with an attention mechanism for time-series quality forecasting from Internet-of-Things (IoT) sensor data; (3) a multimodal feature fusion network that merges visual and temporal embeddings into a unified Digital Twin quality state; (4) an autonomous agent decision module that monitors quality trajectories and generates tiered alerts with actionable recommendations; and (5) a large language model (LLM)-powered dashboard that produces plain-English quality reports for logistics managers. A physics-based frozen degradation simulation grounded in Arrhenius kinetics is introduced as a novel data augmentation strategy to address the absence of public frozen fruit datasets. Experimental results demonstrate that the proposed fusion model achieves a Root Mean Squared Error (RMSE) of 0.0954 on quality score prediction, surpassing the project target of $RMSE < 0.15$, while the CNN classifier achieves above 80% validation accuracy on the three-class quality classification task. The proposed framework represents the first end-to-end system to combine multimodal digital twin fusion, frozen-specific degradation simulation, explainable AI, and agent-based decision making for frozen produce quality management.

Index Terms—digital twin, frozen fruit quality, convolutional neural network, bidirectional LSTM, attention mechanism, cold-chain monitoring, explainable AI, Grad-CAM, Arrhenius degradation model, multimodal fusion, agent-based systems, IoT sensors

I. INTRODUCTION

The global food cold chain is an important part of the safety quality and nutritional integrity of perishable goods from harvest to consumer. Among the frozen, the quality deteriorates rapidly during cold-chain storage insidiously because deterioration occurs internally driven by physical and biochemical mechanisms that are invisible to conventional inspection methods. Ice crystals puncture structures, cellular membranes freeze-thaw cycles caused by temperature variations lead to irreversible moisture migration, sublimation-induced dehydration results in a characteristic grey-white patches of freezer burn [1]. These phenomena collectively absent from fresh-fruit work. They define a distinct and underexplored domain within food science and machine learning literature.

The economic consequences of cold-chain failure are substantial. Global food waste costs exceed \$1 trillion annually, with an estimated 30% of frozen food lost to cold-chain inadequacies including undetected temperature abuse, repeated freeze-thaw events, and prolonged sub-optimal storage [2]. Current monitoring approaches—periodic laboratory analysis, surface visual inspection, or simple threshold-based temperature alarms—are fundamentally reactive: by the time degradation is detectable, interventional opportunity has passed.

Digital twin technology, which constructs continuously-updated virtual replicas of physical systems from real-time sensor data, offers a compelling paradigm for proactive quality management [3]. Agent-based systems further extend this capability by enabling autonomous decision-making based on the digital twin state. However, the convergence of these paradigms with multimodal machine learning for the specific challenge of frozen fruit quality management remains unexplored.

This work addresses this gap by proposing an agent-based digital twin framework that integrates CNN-based visual classification, BiLSTM temporal forecasting, multimodal feature fusion, autonomous agent decision-making, and LLM-powered explainability. The principal contributions of this work are as follows:

- A physics-based frozen degradation simulation grounded in Arrhenius kinetics that generates a novel “slightly

degraded” image class absent from all existing public datasets.

- A multimodal Digital Twin state representation that fuses 1280-dimensional CNN visual embeddings with 256-dimensional BiLSTM temporal embeddings into a unified quality state vector.
- An autonomous agent decision module that generates tiered, actionable alerts from the Digital Twin state, extending prior work beyond passive prediction.
- Grad-CAM explainability applied specifically to frozen fruit degradation imagery, producing human-readable visual justifications for model decisions.

The remainder of this paper is organized as follows. Section II reviews related work. Section III details the proposed methodology.

II. LITERATURE REVIEW

A. Digital Twins for Cold Chain Management

Hertog et al. [1] developed a physics-based digital twin for citrus fruit cold-chain transport, translating sensor data from 47 commercial shipments into quality and marketability metrics using principal component analysis. While the work demonstrated the viability of digital twin methodology in postharvest logistics, it was limited to fresh citrus, employed no machine learning, and included neither image-based quality assessment nor autonomous decision-making. The agricultural digital twin review by Filippi et al. [4] identified post-harvest cold storage of frozen produce as a significantly underexplored area, with reviewed digital twins predominantly addressing field-based crop growth rather than storage-phase quality management.

B. Deep Learning for Fruit Quality Classification

Amin et al. [5] proposed a deep convolutional neural network approach for fruit freshness classification using transfer learning on the Kaggle Fresh and Rotten Fruits dataset. The authors achieved strong binary classification accuracy across apple, banana, and orange categories. However, the work was confined to two quality classes—fresh and rotten—with no intermediate degradation stage modeled. Crucially, all images represented room-temperature fresh fruit, with no frozen-specific visual effects simulated or considered. Similarly, Zanoni et al. [6] applied ResNet-50 transfer learning to fruit quality classification with comparable binary accuracy, but without explainability components, frozen-specific augmentation, or sensor data integration.

C. Time-Series Forecasting for Cold Chain Quality

Liu et al. [7] demonstrated LSTM-based shelf-life prediction for strawberries using temperature and humidity time-series data, achieving shelf-life estimation errors below ± 1.2 days. While this work established the feasibility of recurrent neural networks for postharvest quality forecasting, it was limited to a single fruit type, a single data modality, and did not incorporate freeze-thaw event modeling, bidirectional sequence processing, or attention mechanisms. Jiang et al. [8] combined K-Means++ clustering with LSTM for cold-chain environment

prediction, demonstrating 10-minute temperature forecasting. This work, however, predicted environmental conditions rather than fruit quality scores, and lacked any image channel or decision-making layer.

D. Explainable AI in Food Quality Assessment

A 2025 systematic review of explainable AI techniques in food engineering [9] identified Grad-CAM and SHAP as promising interpretability approaches but confirmed that XAI remains significantly underutilized in food quality research, limiting model reliability and stakeholder trust. No reviewed work applied visual explainability specifically to frozen fruit degradation imagery.

E. Identified Research Gaps

The preceding review reveals a consistent pattern: each prior work addressed one dimension of the problem in isolation. No published study has simultaneously addressed: (1) frozen-specific degradation physics in image augmentation; (2) three-class quality labeling with a simulated intermediate class; (3) multimodal fusion of visual and sensor embeddings; (4) Grad-CAM explainability applied to frozen produce; and (5) autonomous agent-based decision-making integrated with a digital twin quality state. The present work addresses all five gaps within a unified end-to-end pipeline.

III. METHODOLOGY

The proposed framework comprises five sequentially integrated modules. Fig. 1 illustrates the complete system architecture.

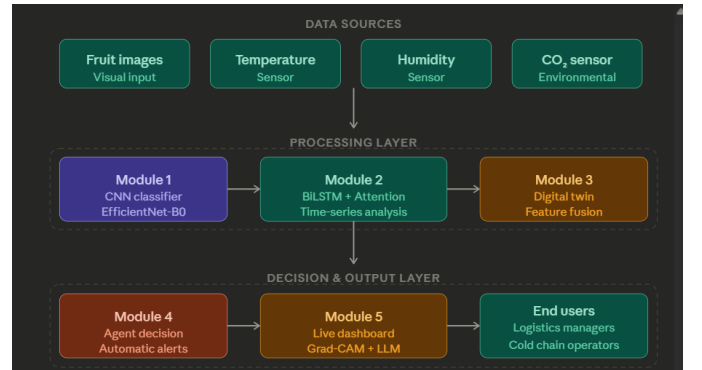


Fig. 1. Agent-based digital twin pipeline. Fruit images and IoT sensor readings feed Module 1 (CNN) and Module 2 (BiLSTM) respectively. Module 3 fuses both embeddings into the Digital Twin state. Module 4 generates autonomous alerts. Module 5 delivers plain-English reports via Claude LLM.

A. Dataset and Novel Frozen Degradation Simulation

1) *Image Dataset*: The Kaggle Fruits Fresh and Rotten for Classification dataset [10] provides 13,599 labeled images across six categories (fresh and rotten variants of apples, bananas, and oranges), pre-divided into training and test splits at approximately 80:20. Images were reorganized into three quality classes: *Fresh* (Class 0), *Slightly Degraded* (Class 1), and *Spoiled* (Class 2).

2) *Novel Frozen Degradation Simulation*: No public dataset captures the visual characteristics specific to frozen fruit degradation. To address this, the *Slightly Degraded* class was generated through physics-based image simulation of three frozen-specific phenomena:

- 1) **Ice crystal formation**: Gaussian blur with random Gaussian frost noise and a blue-white tint overlay to simulate light scattering off ice crystals within fruit tissue.
- 2) **Freeze-thaw cycle damage**: Randomized elliptical dark blotches with Gaussian edge softening to simulate moisture migration and surface darkening from repeated partial thawing and refreezing.
- 3) **Freezer burn**: Circular de-saturation patches with grey-white fill to simulate sublimation-induced surface dehydration.

B. Module 1: CNN Image Classifier

A convolutional neural network (CNN) model is used to classify fruit images into three categories: Fresh, Slightly Degraded and Spoiled. Transfer learning is applied using EfficientNet-B0 to improve performance. Images are pre-processed through resizing, normalization, and augmentation. Explainability is achieved using Grad-CAM, which highlights important regions influencing the prediction.

Grad-CAM [13] was applied to the final convolutional block to generate spatial attention heatmaps, enabling human-interpretable visual explanation of classification decisions.

C. Module 2: BiLSTM Time-Series Forecaster

Sensor data including temperature, humidity, and CO levels are used to model degradation over time. An LSTM model is trained to capture temporal dependencies and predict future quality trends. To address the lack of real-world frozen datasets, synthetic data is generated using freeze-thaw simulation based on physical degradation principles. This includes:

- Temperature fluctuations
- Door opening events
- Partial thaw cycles

These simulations improve model robustness and realism.

D. Module 3: Digital Twin Fusion

Digital Twin Fusion The outputs of the CNN and LSTM models are combined into a unified representation known as the digital twin. This fusion creates a dynamic quality state that reflects both current and predicted conditions. Feature-level fusion techniques are used to integrate embeddings from both models, enabling better prediction accuracy and system stability.

E. Module 4: Agent Decision Module

The agent continuously monitors the Digital Twin quality score $\hat{q}$ and applies a tiered decision policy derived from EDA-defined thresholds:

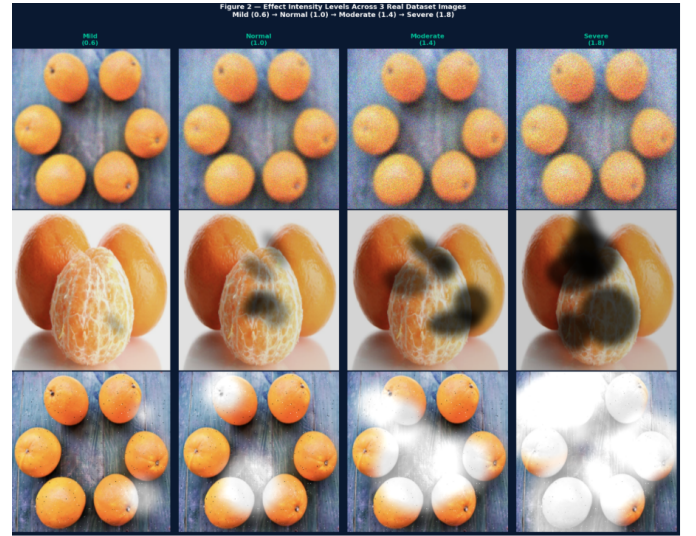
$$\text{Alert}(\hat{q}) = \begin{cases} \text{Normal} & \hat{q} > 0.70 \\ \text{Early Warning} & 0.40 < \hat{q} \leq 0.70 \\ \text{Critical Alert} & \hat{q} \leq 0.40 \end{cases} \quad (1)$$

Distinct from simple threshold-alarm systems, the agent contextualizes decisions using storage day, freeze-thaw event history, and rate of quality decline to generate batch-specific recommendations (e.g., temperature reduction, expedited dispatch, or immediate removal).

F. Module 5: LLM-Powered Dashboard

The Claude Sonnet API (Anthropic) is invoked every 15 minutes with a structured prompt containing the Digital Twin quality state, last 10 sensor readings, agent alert status, and CNN visual classification. Claude generates a four-component plain-English report: quality status, root cause analysis, recommended action, and confidence level. Prompt engineering (zero-shot, no fine-tuning or retrieval-augmented generation) was sufficient given Claude’s pre-existing food science domain knowledge. A sliding window of 10 batch readings provides temporal context without exceeding token limits. Results are rendered on a Streamlit dashboard alongside the 90-day quality forecast chart, active alerts, and Grad-CAM heatmaps.

IV. TEST EVALUATION



Each row shows a real fruit image being analyzed by four different degradation effects, ice crystals spread noise evenly across the whole surface, freeze-thaw creates concentrated bright spots where moisture damage occurred, freezer burn leaves large soft circular patches where dehydration happened, and the combined column merges two effects together to create the slightly degraded class used in our CNN training.

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